Task 1: Enable AWS GuardDuty

Procedure
1. Sign in to AWS.
2. Search for Amazon GuardDuty.
3. Open the GuardDuty console.
4. Select the appropriate AWS Region.
5. Choose Get started or Enable GuardDuty, depending on the console view.
6. Confirm that GuardDuty is enabled.
7. Review the dashboard and findings page.

Task 2: Test suspicious login activity safely

Repeated incorrect console passwords do not necessarily generate a GuardDuty finding. GuardDuty primarily detects threats using supported cloud telemetry, and failed console authentication is better investigated through AWS sign-in events and related monitoring.

A safer test is to use an authorized test IAM identity and deliberately perform a denied API action, such as attempting an operation for which the identity has no permission.

Example:

```
aws s3api create-bucket \
  --bucket my-unauthorized-test-bucket-unique-id \
  --region ap-south-1
```

Task 3: Explore Microsoft Defender for Cloud

Microsoft has renamed Azure Security Center to Microsoft Defender for Cloud.

Procedure

1. Sign in to the Azure portal.
2. Search for Microsoft Defender for Cloud.
3. Open Recommendations and Security alerts.
4. Review the security posture of the available subscription and resources.
5. If you have a test virtual machine, review its protection status and applicable recommendations.
6. Enable a Defender plan only if required for the lab and after reviewing the cost and trial eligibility.

Task 4: Compare AWS GuardDuty and Microsoft Defender for Cloud

AWS GuardDuty	Microsoft Defender for Cloud
Detects potentially compromised AWS credentials and suspicious API activity.	Identifies security alerts and recommendations across supported Azure and connected resources.
Detects unusual network behavior and potential communication with malicious infrastructure.	Detects threats involving supported virtual machines, workloads, and other protected resources.
Identifies potential reconnaissance, suspicious access patterns, and certain malicious workload activities.	Highlights vulnerabilities, insecure configurations, and supported workload threats.

Task 5: How real-time threat detection works

Example: A potential brute-force attack against a cloud resource.
1. Data collection: Cloud services generate authentication events, API activity, and relevant network telemetry.

2. Analysis: The security service evaluates supported data against threat intelligence, behavioral patterns, and detection rules.
3. Detection: Suspicious patterns may result in a finding or security alert.
4. Investigation: The administrator checks the source, timestamps, affected identity or resource, and related events.
5. Response: The team can revoke compromised credentials, require stronger authentication, restrict network access, or isolate a compromised workload.
6. Verification: The administrator confirms that the suspicious activity has stopped and records the incident.